

CodeCompass

AI Career Platform

PROJECT TYPE AI Career Platform	PLATFORM Web and native Android applications
AUDIENCE Philippine CS students and early-career developers	PRIMARY FOCUS AI-guided technology career planning

Overview

An AI-driven career guidance platform built for Computer Science students in the Philippines, shipped on both Web and Android. CodeCompass generates personalized learning roadmaps, surfaces local IT jobs and CCS programs, tracks progress through XP and badges, and embeds an AI Career Coach for resume help and interview prep.

Core Features

- AI-generated personalized career roadmaps with skill nodes and XP-gated progression
- AI Career Coach with multi-session chat for resume and interview prep
- Live PH job listings (Careerjet, JSearch) with resume-based recommendations
- Resume Builder with three templates, AI content tools, and ATS scoring
- Certifications tracker (TESDA, Google, AWS) with progress sync
- Philippine university finder with CHED-rated CCS programs
- Gamified achievements: XP, daily streaks, badges, and leaderboards
- Cross-platform: React + Vite web and Java + Material 3 Android on a shared Django REST + Channels backend

Experience and Design

- Personalized roadmaps break career goals into skills, resources, and XP-gated milestones.
- An AI Career Coach supports resume writing, interview preparation, and career questions.
- Job, certification, and university discovery connect learning plans with local opportunities.
- Achievements, streaks, badges, and leaderboards make long-term progress visible.

Architecture and Data

- React and Vite deliver the browser application, while Java and Material Design 3 power Android.
- A shared Django REST backend serves profiles, roadmaps, jobs, resumes, and achievement data.
- Django Channels and WebSocket connections stream AI responses and real-time updates.
- Background services support scheduled data refreshes and longer-running recommendation tasks.

Security and Privacy

- JWT access and refresh tokens protect authenticated API requests.
- Android stores tokens in encrypted preferences rather than plain local storage.
- Email verification, rate limiting, and server-side validation protect sensitive workflows.
- Roadmaps, chats, resumes, saved jobs, and progress records are isolated per account.

Technology Stack

React 18	Application framework and UI
Vite	Application framework and UI
Tailwind CSS	Interface styling and design system
Java 11	Application language
Material Design 3	Interface styling and design system
Django REST	Backend application services
Groq SDK	AI and speech processing
WebSocket	Real-time communication

Performance and Reliability

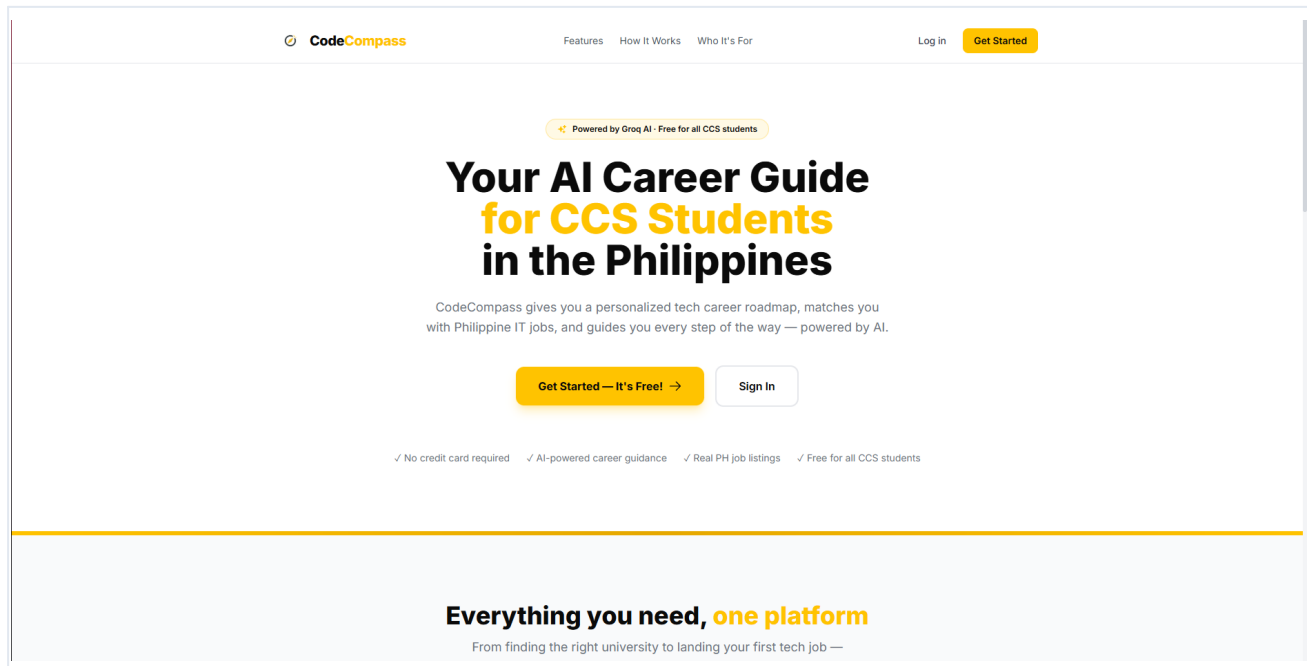
- WebSocket streaming avoids repeated polling during AI conversations.
- Cached reference data reduces repeated job, certification, and university requests.
- Background workers handle scheduled and compute-heavy tasks outside request cycles.
- Shared API models keep web and Android progress synchronized efficiently.

Platform Requirements

- Current Chrome, Edge, Firefox, or Safari for the web application.
- Supported Android device for the native application.
- Stable internet connection for AI, live jobs, and synchronized progress.
- Account sign-in is required for personalized recommendations and saved work.

PAGE OVERVIEW

Landing Page (Web)

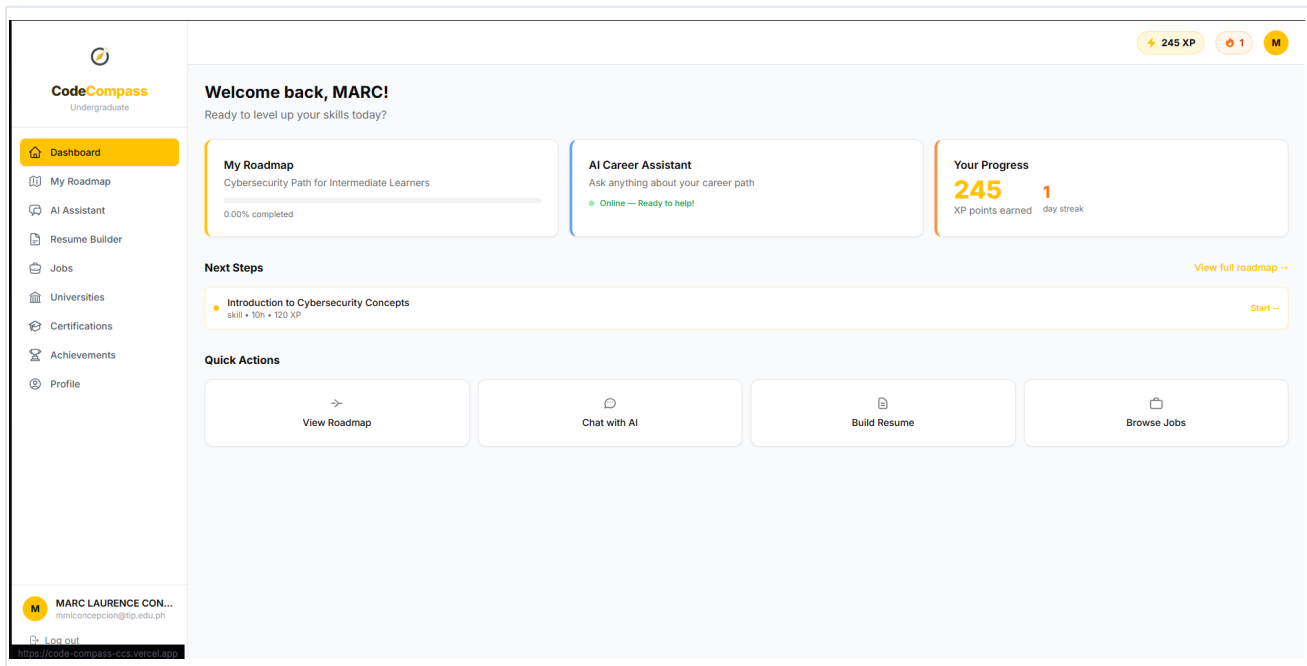


Landing Page (Web)

Interface view from CodeCompass.

PAGE OVERVIEW

Dashboard (Web)

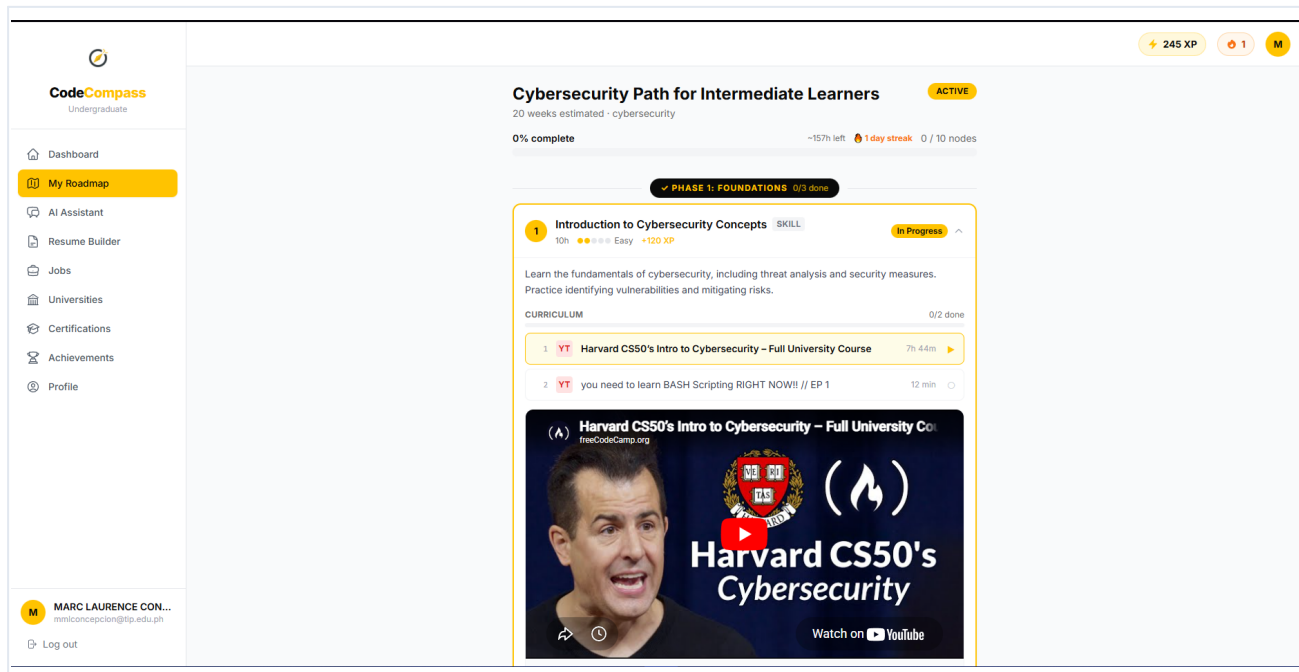


Dashboard (Web)

Interface view from CodeCompass.

PAGE OVERVIEW

Career Roadmap (Web)

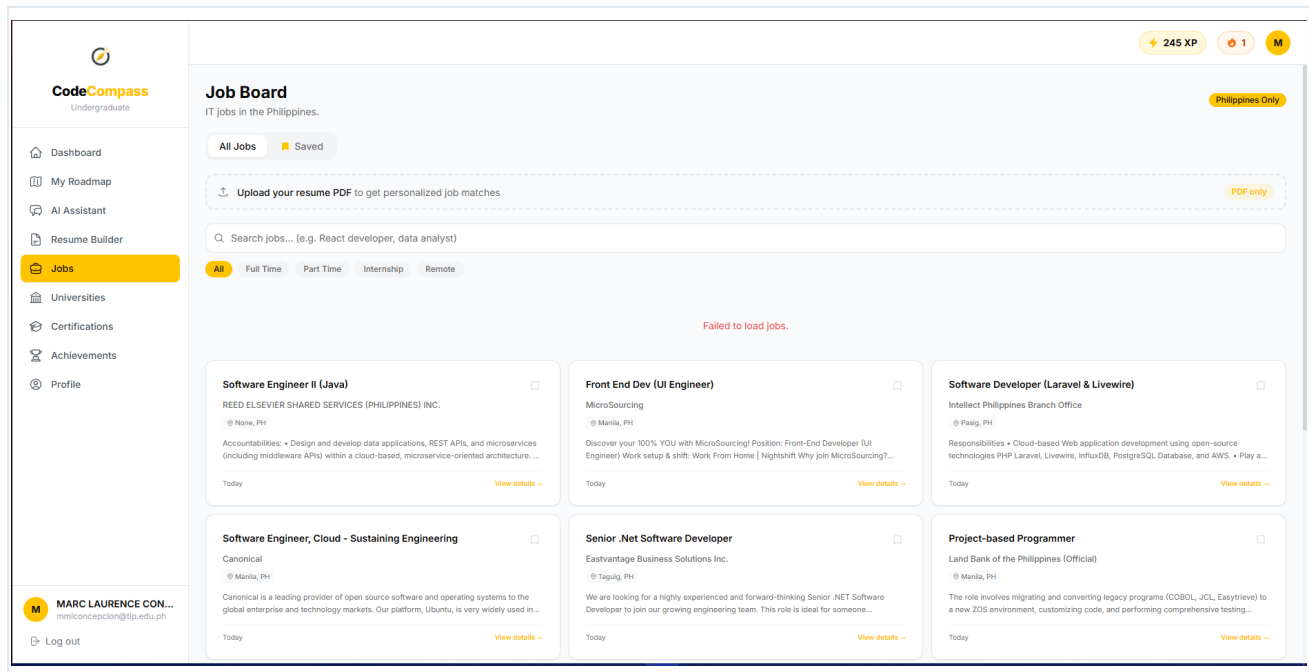


Career Roadmap (Web)

Interface view from CodeCompass.

PAGE OVERVIEW

Job Board (Web)

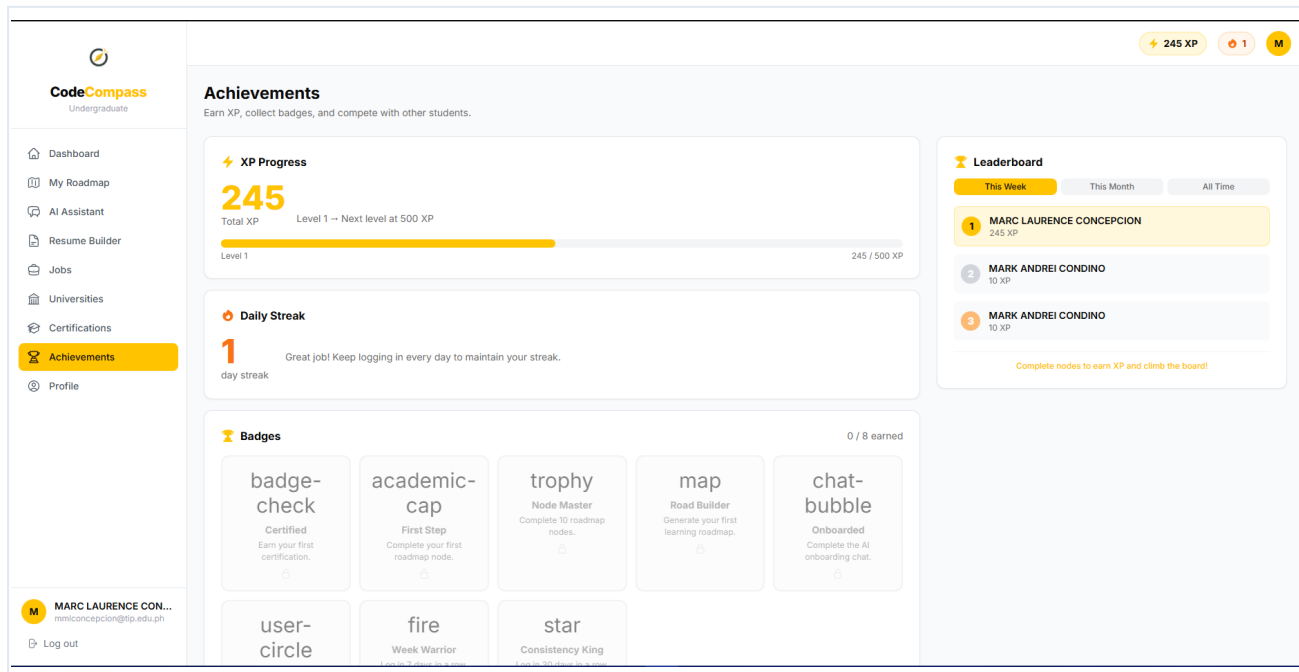


Job Board (Web)

Interface view from CodeCompass.

PAGE OVERVIEW

Achievements (Web)

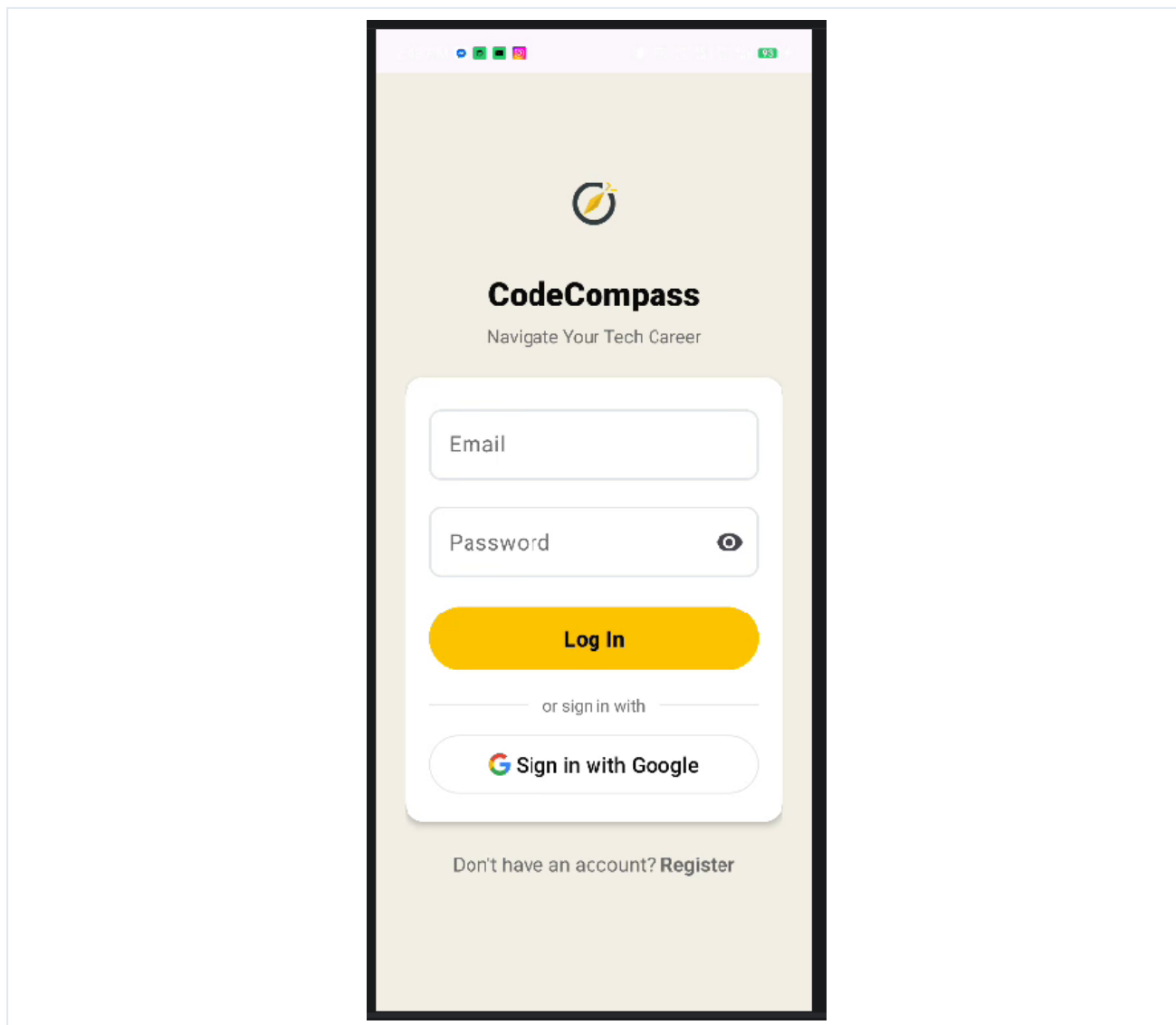


Achievements (Web)

Interface view from CodeCompass.

PAGE OVERVIEW

Login (Android)

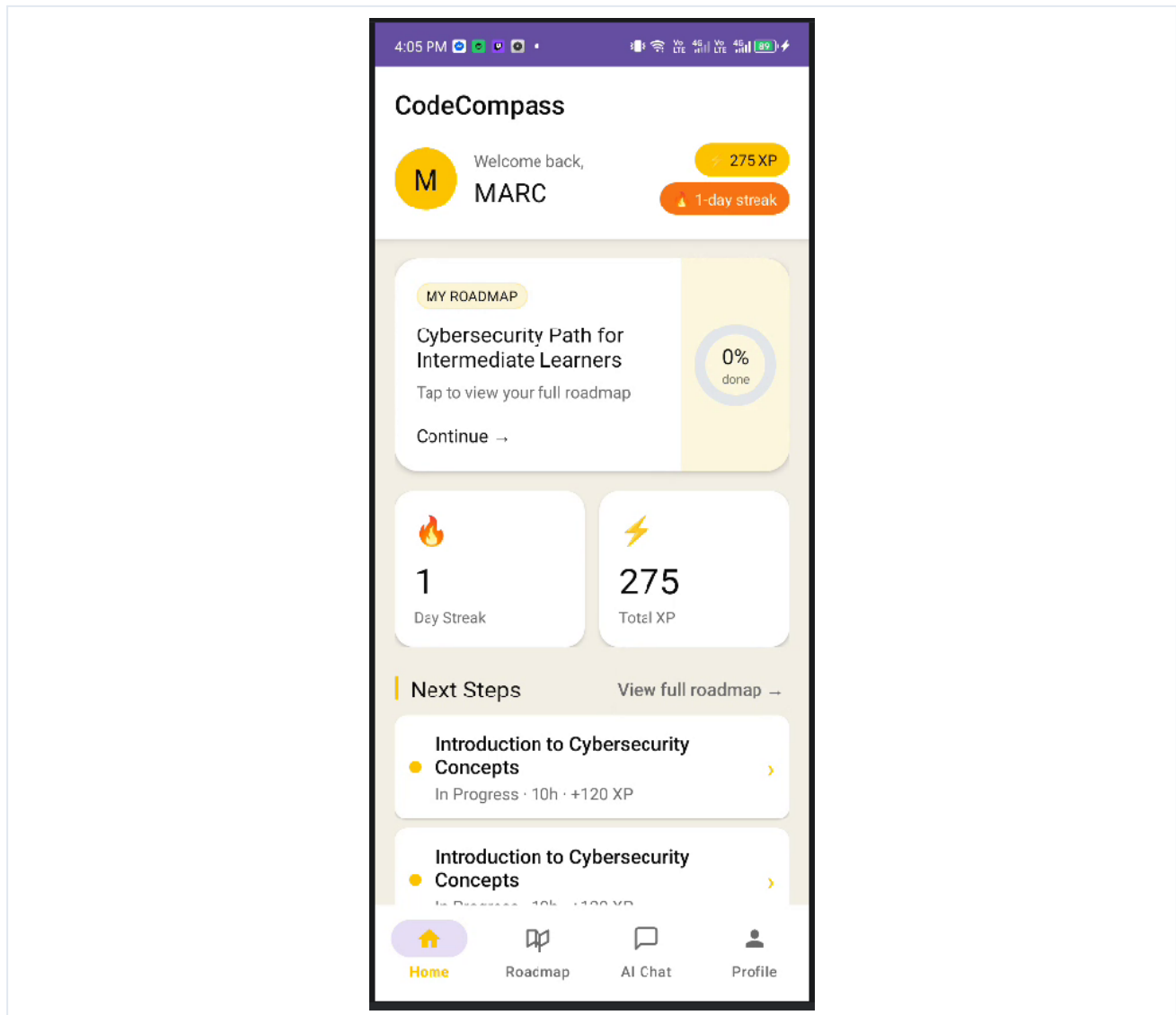


Login (Android)

Interface view from CodeCompass.

PAGE OVERVIEW

Home (Android)

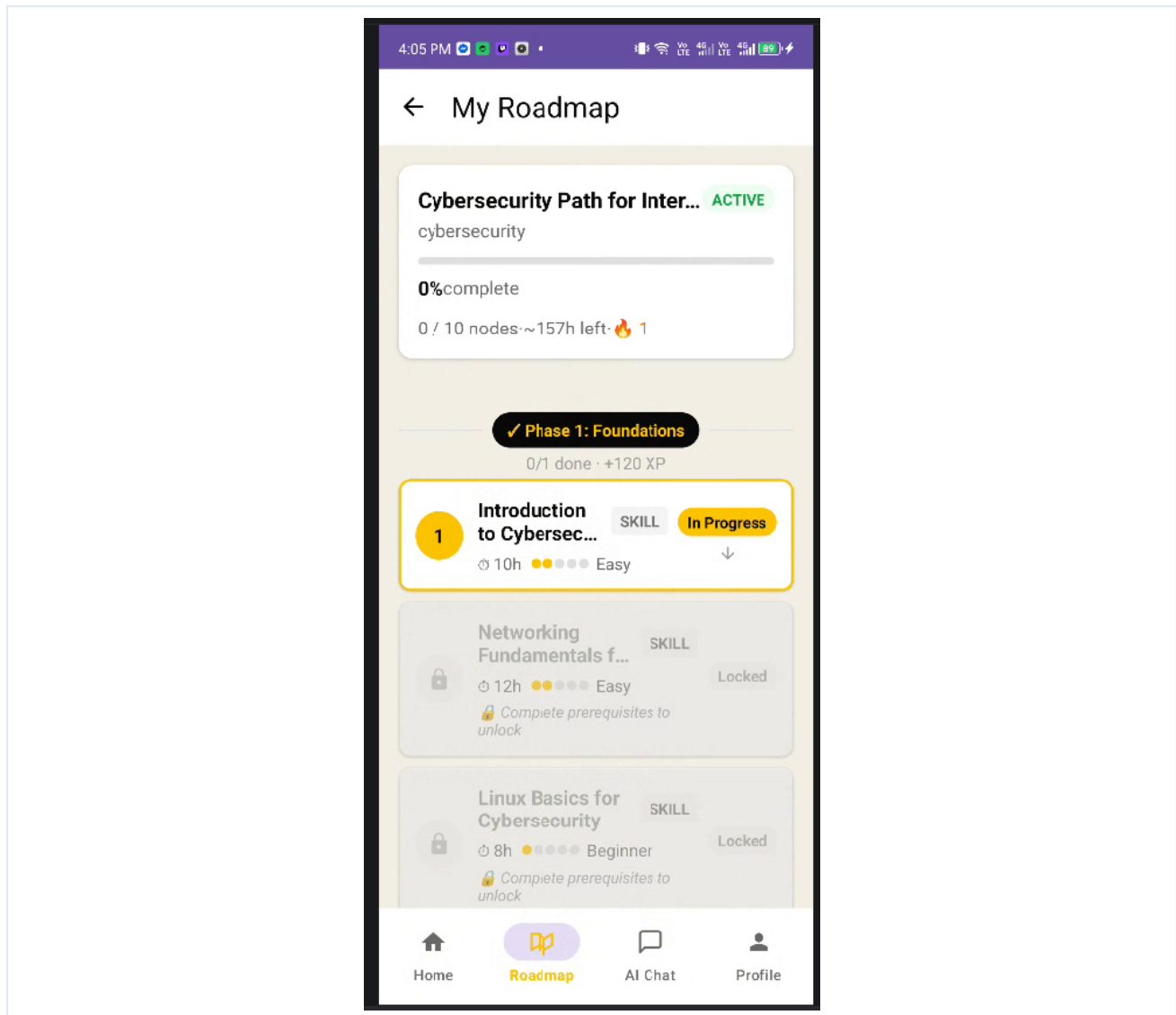


Home (Android)

Interface view from CodeCompass.

PAGE OVERVIEW

Career Roadmap (Android)

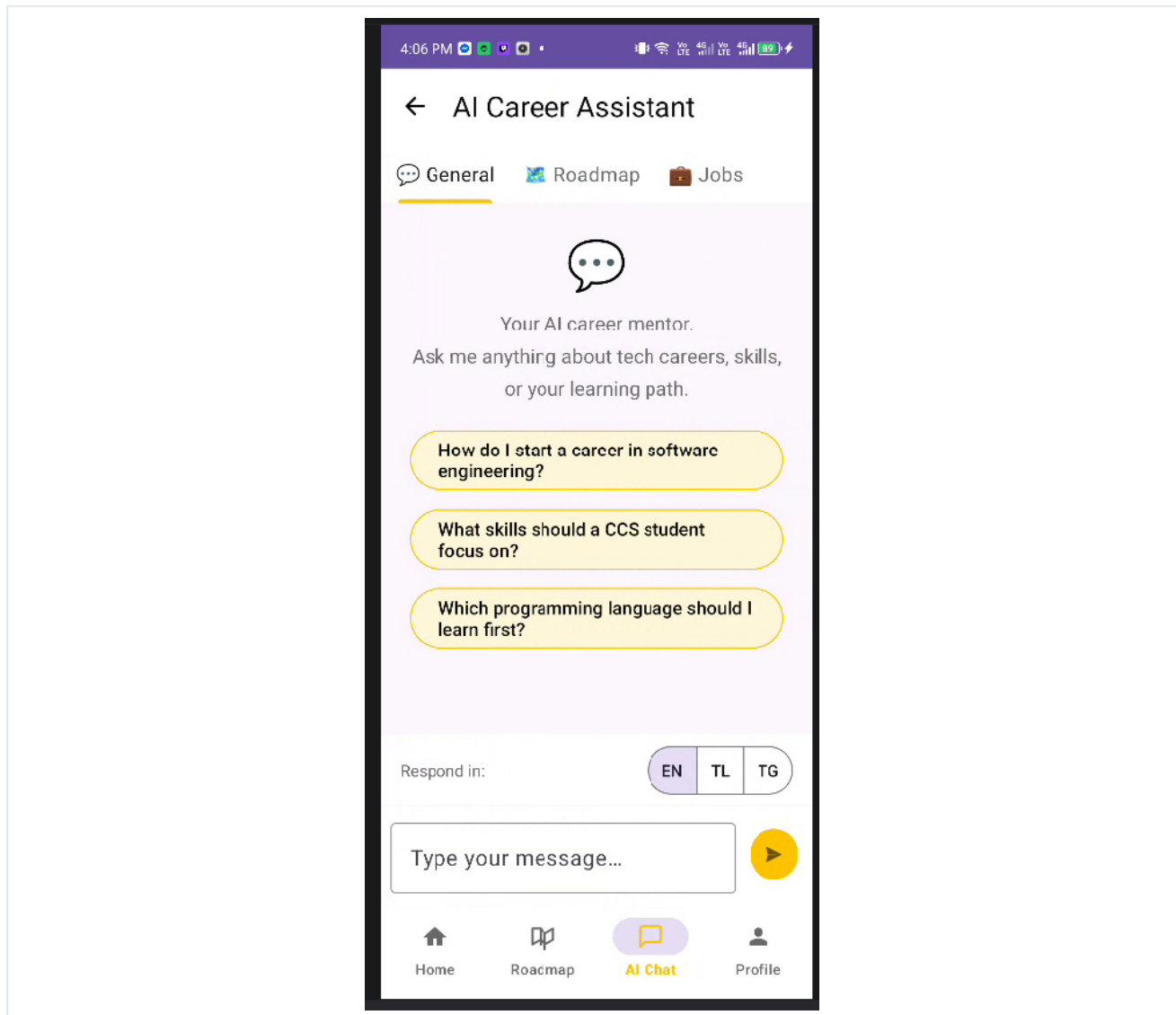


Career Roadmap (Android)

Interface view from CodeCompass.

PAGE OVERVIEW

AI Career Coach (Android)

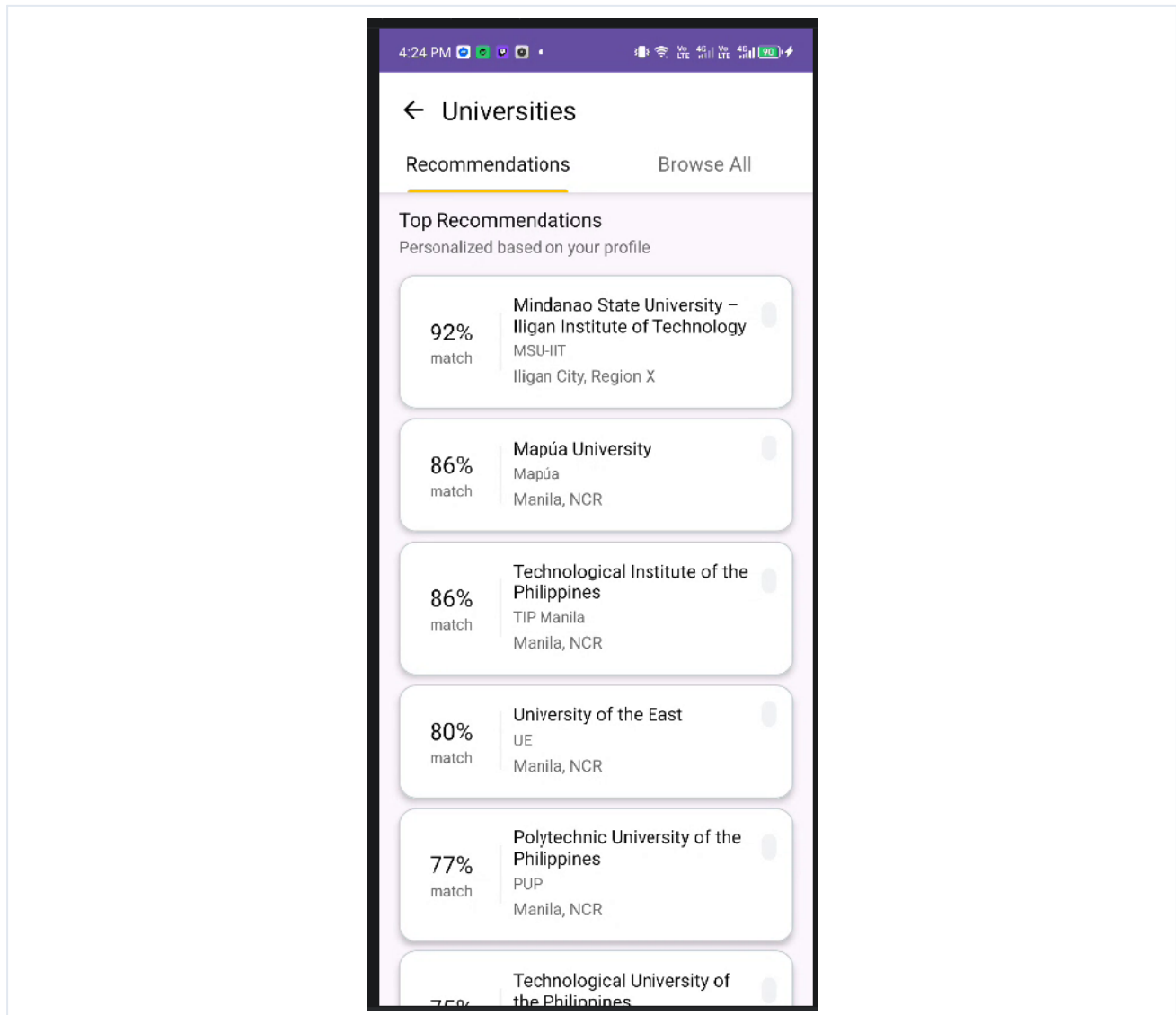


AI Career Coach (Android)

Interface view from CodeCompass.

PAGE OVERVIEW

University Finder (Android)



University Finder (Android)

Interface view from CodeCompass.